Exact Differential Equation

Perfect differential:

Let M and N both are functions of x, y. If $Mdx + Ndy = 0$ is obtained by differentiating $f(x, y) = c$ then $Mdx + Ndy = 0$ is called perfect or exact differential.

Example: $xy = c$ or, $ydx + xdy = 0$

Exact Differential Equation:

If any differential equation has exact differential then this is called exact differential equation.

The condition of exactness of the DE $Mdx + Ndy = 0$ is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Integrating Factor:

A DE is multiplied by a factor and then the equation becomes integrable , such factor is called integrating factor. It is denoted by IF.A non-exact DE becomes exact after multiplying it by IF.

Solution:

If $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ i,e, the DE is exact then its solution is $\int_{y\ constant} Mdx + \int N(term\ free\ from\ x)dy = c$

If $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ i,e, the DE is not exact then

Rule-I: If M and N are homogeneous and If $Mx + Ny \neq 0$ then $IF = \frac{1}{Mx + Ny}$

Rule-II: If the equation is of the form $f(xy)ydx + \phi(xy)xdy = 0$ then if $Mx - Ny \neq 0$ then $IF = \frac{1}{Mx - Ny}$

Rule-III: If $\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = f(x)$ [i,e, a function of x only] then $IF = e^{f(x)dx}$

Rule-IV: If $\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = f(y)$ [i,e, a function of y only] then $IF = e^{f(y)dy}$

Ex: Solve $(3x + 2y - 5)dx + (2x + 3y - 5)dy = 0$

Solu: Given equation is $(3x + 2y - 5)dx + (2x + 3y - 5)dy = 0$

$Here,\ M = 3x + 2y - 5 \quad \therefore \frac{\partial M}{\partial y} = 2 \qquad N = 2x + 3y - 5 \quad \therefore \frac{\partial N}{\partial x} = 2$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is $\int_{y\ constant} Mdx + \int N(term\ free\ from\ x)dy = c$

$or,\ \int_{y\ constant} (3x + 2y - 5)dx + \int (3y - 5)dy = c_1 \qquad or,\ 3\int xdx + 2y\int dx - 5\int dx + 3\int ydy - 5\int dy = c_1$

$or,\ \frac{3x^2}{2} + 2xy - 5x + \frac{3y^2}{2} - 5y = c_1 \qquad or,\ 3x^2 + 4xy - 10x + 3y^2 - 10y = c$

This is the required solution

Ex: Solve $(4x + 3y + 1)dx + (3x + 2y + 1)dy = 0$

Solu: Given equation is $(4x + 3y + 1)dx + (3x + 2y + 1)dy = 0$

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Here, $M = 4x + 3y + 1 \quad \therefore \frac{\partial M}{\partial y} = 3 \qquad N = 3x + 2y + 1 \quad \therefore \frac{\partial N}{\partial x} = 3$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is $\int\limits_{y\ cons\tan t} Mdx + \int N(term\ free\ from\ x)dy = c$

or, $\int\limits_{y\ cons\tan t} (4x + 3y + 1)dx + \int (2y + 1)dy = c \qquad or, \quad 2x^2 + 3xy + x + y^2 + y = c$

This is the required solution

Ex: Solve $(2x + 3y + 4)dx + (3x + 4y + 5)dy = 0$

Solu: Given equation is $(2x + 3y + 4)dx + (3x + 4y + 5)dy = 0$

Here, $M = 2x + 3y + 4 \quad \therefore \frac{\partial M}{\partial y} = 3 \qquad N = 3x + 4y + 5 \quad \therefore \frac{\partial N}{\partial x} = 3$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so the differential equation is exact.

$\therefore$ solution is $\int\limits_{y\ cons\tan t} Mdx + \int N(term\ free\ from\ x)dy = c$

or, $\int\limits_{y\ cons\tan t} (2x + 3y + 4)dx + \int (4y + 5)dy = c \qquad or, \quad x^2 + 3xy + 4x + 2y^2 + 5y = c$

This is the required solution

Ex: Solve $(\sin x\cos y + e^{2x})dx + (\cos x\sin y + \tan y)dy = 0$

Solu: Given equation is

$(\sin x\cos y + e^{2x})dx + (\cos x\sin y + \tan y)dy = 0$

Here

$M = \sin x\cos y + e^{2x} \quad \therefore \frac{\partial M}{\partial y} = -\sin x\sin y \qquad N = \cos x\sin y + \tan y \quad \therefore \frac{\partial N}{\partial x} = -\sin x\sin y$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is $\int\limits_{y\ cons\tan t} Mdx + \int N(term\ free\ from\ x)dy = c$

or, $\int\limits_{y\ cons\tan t} (\sin x\cos y + e^{2x})dx + \int \tan ydy = c_1 \qquad or, \quad \cos y\int \sin xdx + \int e^{2x}dx + \int \tan ydy = c_1$

or, $-\cos x\cos y + \frac{e^{2x}}{2} + \ln(\sec y) = c_1 \qquad or, \quad -2\cos x\cos y + e^{2x} + 2\ln(\sec y) = c$

This is the required solution

Ex: Solve $(2xy + y - \tan y)dx + (x^2 - x\tan^2 y + \sec^2 y)dy = 0$

Solu: Given equation is $(2xy + y - \tan y)dx + (x^2 - x\tan^2 y + \sec^2 y)dy = 0$

Here

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$M = 2xy + y - \tan y \qquad \therefore \dfrac{\partial M}{\partial y} = 2x + 1 - \sec^2 y = 2x - \tan^2 y$

$N = x^2 - x\tan^2 y + \sec^2 y \qquad \therefore \dfrac{\partial N}{\partial x} = 2x - \tan^2 y$

Since $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is $\displaystyle\int_{y\ constant} Mdx + \int N(term\ free\ from\ x)dy = c$

$or,\ \displaystyle\int_{y\ constant} (2xy + y - \tan y)dx + \int \sec^2 y dy = c$

$or,\ 2y\displaystyle\int xdx + y\int dx - \tan y\int dx + \int \sec^2 y dy = c$

$or,\ x^2 y + xy - x\tan y + \tan y = c$

This is the required solution

Ex: Solve $(x^3 + 3xy^2)dx + (y^3 + 3x^2 y)dy = 0$

Solu: Given equation is $(x^3 + 3xy^2)dx + (y^3 + 3x^2 y)dy = 0$

Here $M = x^3 + 3xy^2 \qquad \therefore \dfrac{\partial M}{\partial y} = 6xy \qquad N = y^3 + 3x^2 y \qquad \therefore \dfrac{\partial N}{\partial x} = 6xy$

Since $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is $\displaystyle\int_{y\ constant} Mdx + \int N(term\ free\ from\ x)dy = c$

$or,\ \displaystyle\int_{y\ constant} (x^3 + 3xy^2)dx + \int y^3 dy = c_1 \qquad or,\ \int x^3 dx + 3y^2\int xdx + \int y^3 dy = c_1$

$or,\ \dfrac{x^4}{4} + \dfrac{3x^2y^2}{2} + \dfrac{y^4}{4} = c_1 \qquad or,\ x^4 + 6x^2y^2 + y^4 = c$

This is the required solution

Ex: Solve $(y^4 + 4x^3 y + 3x)dx + (x^4 + 4xy^3 + y + 1)dy = 0$

Solu: Given equation is $(y^4 + 4x^3 y + 3x)dx + (x^4 + 4xy^3 + y + 1)dy = 0$

Here

$M = y^4 + 4x^3 y + 3x \qquad \therefore \dfrac{\partial M}{\partial y} = 4y^3 + 4x^3 \qquad N = x^4 + 4xy^3 + y + 1 \qquad \therefore \dfrac{\partial N}{\partial x} = 4x^3 + 4y^3$

Since $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is $\displaystyle\int_{y\ constant} Mdx + \int N(term\ free\ from\ x)dy = c$

$or,\ \displaystyle\int_{y\ constant} (y^4 + 4x^3 y + 3x)dx + \int (y+1)dy = c_1 \qquad or,\ y^4\int dx + 4y\int x^3 dx + 3\int xdx + \int (y+1)dy = c_1$

$or,\ xy^4 + 4y.\dfrac{x^4}{4} + 3\dfrac{x^2}{2} + \dfrac{y^2}{2} + y = c_1 \qquad or,\ 2xy^4 + 2x^4 y + 3x^2 + y^2 + 2y = c$

This is the required solution

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Ex: Solve $(x^2-2xy+3y^2)dx+(4y^3+6xy-x^2)dy=0$

Solu: Given equation is $(x^2-2xy+3y^2)dx+(4y^3+6xy-x^2)dy=0$

Here $M=x^2-2xy+3y^2$ $\quad\therefore \dfrac{\partial M}{\partial y}=-2x+6y$ $\quad N=4y^3+6xy-x^2$ $\quad\therefore \dfrac{\partial N}{\partial x}=6y-2x$

Since $\dfrac{\partial M}{\partial y}=\dfrac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is

$$\int_{y\ cons\tan t} Mdx+\int N(term\ free\ from\ x)dy=c$$

$$or,\quad \int_{y\ cons\tan t}(x^2-2xy+3y^2)dx+\int 4y^3dy=c_1 \quad or,\quad \int x^2dx-2y\int xdx+3y^2\int dx+4\int y^3dy=c_1$$

$$or,\quad \frac{x^3}{3}-2y.\frac{x^2}{2}+3y^2.x+4.\frac{y^4}{4}=c_1 \qquad or,\quad x^3-3x^2y+9xy^2+3y^4=c$$

This is the required solution

Ex: Solve $(\tan y-3x^2)dx+x\sec^2 ydy=0$

Solu: Given equation is $(\tan y-3x^2)dx+x\sec^2 ydy=0$

Here $M=\tan y-3x^2$ $\quad\therefore \dfrac{\partial M}{\partial y}=\sec^2 y$ $\quad N=x\sec^2 y$ $\quad\therefore \dfrac{\partial N}{\partial x}=\sec^2 y$

Since $\dfrac{\partial M}{\partial y}=\dfrac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is $\displaystyle\int_{y\ cons\tan t} Mdx+\int N(term\ free\ from\ x)dy=c$

$$or,\quad \int_{y\ cons\tan t}(\tan y-3x^2)dx+\int 0dy=c \qquad or,\quad x\tan y-x^3=c$$

This is the required solution

Ex: Solve $(x^2-2xy^2+y^4)dx-(2x^2y-4xy^3+\sin y)dy=0$

Solu: Given equation is $(x^2-2xy^2+y^4)dx-(2x^2y-4xy^3+\sin y)dy=0$

Here

$M=x^2-2xy^2+y^4$ $\quad\therefore \dfrac{\partial M}{\partial y}=-4xy+4y^3$ $\quad N=2x^2y-4xy^3+\sin y$ $\quad\therefore \dfrac{\partial N}{\partial x}=4y^3-4xy$

Since $\dfrac{\partial M}{\partial y}=\dfrac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is $\displaystyle\int_{y\ cons\tan t} Mdx+\int N(term\ free\ from\ x)dy=c$

$$or,\quad \int_{y\ cons\tan t}(x^2-2xy^2+y^4)dx-\int \sin ydy=c_1$$

$$or,\quad \frac{x^3}{3}-2y^2.\frac{x^2}{2}+xy^4+\cos y=c_1$$

$$or,\quad x^3-3x^2y^2+3xy^4+3\cos y=c$$

This is the required solution

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Ex: Solve $(6xy^2+4x^3)dx+(6x^2y+3y^3)dy=0$

Solu: Given equation is $(6xy^2+4x^3)dx+(6x^2y+3y^3)dy=0$

Here

$M=6xy^2+4x^3 \qquad \therefore \frac{\partial M}{\partial y}=12xy \qquad N=6x^2y+3y^3 \qquad \therefore \frac{\partial N}{\partial x}=12xy$

Since $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$ so the differential equation is exact.

$\therefore$ solution is

$$\int_{y\ constant} Mdx+\int N(term\ free\ from\ x)dy=c$$

$$or,\quad \int_{y\ constant}(6xy^2+4x^3)dx-\int 3y^3dy=c$$

$$or,\quad 3x^2y^2+x^4+\frac{3y^4}{4}=c$$

This is the required solution

Ex: Solve $xydx-(x^2+y^2)dy=0$

Solu: Given equation is

$xydx-(x^2+y^2)dy=0$ .....(i)

Here

$M=xy \qquad \therefore \frac{\partial M}{\partial y}=-x \qquad N=-(x^2+y^2) \qquad \therefore \frac{\partial N}{\partial x}=-2x$

Since $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

Now,

$Mx+Ny=xy.x-(x^2+y^2).y=x^2y-x^2y-y^3=-y^3$

$\therefore \quad IF=-\frac{1}{y^3}$

$(i)\times IF \Rightarrow$

$$-\frac{xy}{y^3}dx-\frac{(x^2+y^2)}{-y^3}dy=0 \qquad or,\quad -\frac{x}{y^2}dx+\left(\frac{x^2}{y^3}+\frac{1}{y}\right)dy=0$$

$\therefore$ solution is $\int_{y\ constant} Mdx+\int N(term\ free\ from\ x)dy=c$

$$or,-\int_{y\ const}\frac{x}{y^2}dx+\int\frac{1}{y}dy=c \quad or,\quad -\frac{x^2}{2y^2}+\ln y=c$$

This is the required solution

Ex: Solve $y^2dx+(x^2-xy)dy=0$

Solu: Given equation is

$y^2dx+(x^2-xy)dy=0$ .....(i)

Here

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$M = y^2 \qquad \therefore \frac{\partial M}{\partial y} = 2y \qquad N = x^2 - xy \qquad \therefore \frac{\partial N}{\partial x} = 2x - y$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

Now,

$Mx + Ny = y^2.x + (x^2 - xy).y = xy^2 + x^2y - y^3 - xy^2 = x^2y$

$\therefore \quad IF = \frac{1}{x^2y}$

$(i) \times IF \Rightarrow$

$$\frac{y^2}{x^2y}dx + \frac{(x^2 - xy)}{x^2y}dy = 0 \qquad or, \quad \frac{y}{x^2}dx + \left(\frac{1}{y} - \frac{1}{x}\right)dy = 0$$

$\therefore$ solution is $\int_{y\ const} Mdx + \int N(term\ free\ from\ x)dy = c$

$$or, \quad \int_{y\ const} \frac{y}{x^2}dx + \int \frac{1}{y}dy = c$$

$$or, \quad -\frac{y}{x} + \ln y = c$$

This is the required solution

Ex: Solve $y^2(ydx + 2xdy) - x^2(2ydx + xdy) = 0$

Solu: Given equation is

$$y^2(ydx + 2xdy) - x^2(2ydx + xdy) = 0$$

$$or, \quad y^3dx + 2xy^2dy - 2x^2ydx - x^3dy = 0$$

$$or, \quad (y^3 - 2x^2y)dx + (2xy^2 - x^3)dy = 0 \text{ .....(i)}$$

Here

$M = y^3 - 2x^2y \qquad \therefore \frac{\partial M}{\partial y} = 3y^2 - 2x^2 \qquad N = 2xy^2 - x^3 \qquad \therefore \frac{\partial N}{\partial x} = 2y^2 - 3x^2$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

Now,

$Mx + Ny = (y^3 - 2x^2y)x + (2xy^2 - x^3)y = xy^3 - 2x^3y + 2xy^3 - x^3y = 3xy^3 - 3x^3y = 3xy(y^2 - x^2) \neq 0$

$\therefore \quad IF = \frac{1}{3xy(y^2 - x^2)}$

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$(i)\times IF \Rightarrow$

$$\frac{y^3-2x^2y}{3xy(y^2-x^2)}dx+\frac{2xy^2-x^3}{3xy(y^2-x^2)}dy=0$$

$$or,\quad \frac{y(y^2-2x^2)}{xy(y^2-x^2)}dx+\frac{x(2y^2-x^2)}{xy(y^2-x^2)}dy=0$$

$$or,\quad \frac{y^2-x^2-x^2}{x(y^2-x^2)}dx+\frac{y^2-x^2+y^2}{y(y^2-x^2)}dy=0$$

$$or,\quad \left(\frac{y^2-x^2}{x(y^2-x^2)}-\frac{x^2}{x(y^2-x^2)}\right)dx+\left(\frac{y^2-x^2}{y(y^2-x^2)}+\frac{y^2}{y(y^2-x^2)}\right)dy=0$$

$$or,\quad \left(\frac{1}{x}-\frac{x}{y^2-x^2}\right)dx+\left(\frac{1}{y}+\frac{y}{y^2-x^2}\right)dy=0$$

Here,

$$M=\frac{1}{x}-\frac{x}{y^2-x^2}\qquad N=\frac{1}{y}+\frac{y}{y^2-x^2}$$

$\therefore$ solution is

$$\int_{y\,constant} Mdx+\int N(term\ free\ from\ x)dy=c$$

$$or,\quad \int_{y\,const}\left(\frac{1}{x}-\frac{x}{y^2-x^2}\right)dx+\int\frac{1}{y}dy=\ln c$$

$$or,\quad \ln x+\frac{1}{2}\ln(y^2-x^2)+\ln y=\ln c$$

$$or,\quad 2\ln xy+\ln(y^2-x^2)=\ln c$$

$$or,\quad \ln x^2y^2(y^2-x^2)=\ln c$$

$$or,\quad x^2y^2(y^2-x^2)=c$$

This is the required solution

Ex: Solve $(x^2y+y^3)dx-(x^3+2xy^2)dy=0$

Solu: Given equation is

$(x^2y+y^3)dx-(x^3+2xy^2)dy=0$ .....(i)

Here

$$M=x^2y+y^3\qquad \therefore\frac{\partial M}{\partial y}=x^2+3y^2\qquad N=-(x^3+2xy^2)\qquad \therefore\frac{\partial N}{\partial x}=3x^2+2y^2$$

Since $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

Now,

$$Mx+Ny=(x^2y+y^3)x-(x^3+2xy^2)y=x^3y+xy^3-x^3y-2xy^3=-xy^3$$

$$\therefore\ IF=-\frac{1}{xy^3}$$

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$(i)\times IF \Rightarrow$

$$\frac{x^2y+y^3}{-xy^3}dx-\frac{(x^3+2xy^2)}{-xy^3}dy=0$$

$$or,\quad -\left(\frac{x}{y^2}+\frac{1}{x}\right)dx+\left(\frac{x^2}{y^3}+\frac{2}{y}\right)dy=0$$

$\therefore$ solution is

$$\int_{y\ constant} Mdx+\int N(term\ free\ from\ x)dy=c$$

$$or,-\int_{y\ const}\left(\frac{x}{y^2}+\frac{1}{x}\right)dx+\int\frac{2}{y}dy=c$$

$$or,\left(\frac{x^2}{2y^2}+\ln x\right)+2\ln y=c$$

This is the required solution

Ex: Solve $(xy+2x^2y^2)ydx+(xy-x^2y^2)xdy=0$

Solu: Given equation is

$(xy+2x^2y^2)ydx+(xy-x^2y^2)xdy=0$ .....(i)

Here

$M=(xy+2x^2y^2)y=xy^2+2x^2y^3 \qquad \therefore \frac{\partial M}{\partial y}=2xy+6x^2y^2$

$N=(xy-x^2y^2)x=x^2y-x^3y^2 \qquad \therefore \frac{\partial N}{\partial x}=2xy-3x^2y^2$

Since $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$ so the differential equation is not exact. But the equation is of the form

$f(xy)ydx+\phi(xy)xdy=0$

Now, $Mx-Ny=(xy^2+2x^2y^3)x-(x^2y-x^3y^2)y=x^2y^2+2x^3y^3-x^2y^2+x^3y^3=3x^3y^3$

$$\therefore\ IF=\frac{1}{3x^3y^3}$$

$(i)\times IF \Rightarrow$

$$\frac{xy^2+2x^2y^3}{3x^3y^3}+\frac{x^2y-x^3y^2}{3x^3y^3}=0 \quad or,\quad \left(\frac{1}{x^2y}+\frac{2}{x}\right)dx+\left(\frac{1}{xy^2}-\frac{1}{y}\right)dy=0$$

Here, $M=\frac{1}{x^2y}+\frac{2}{x} \quad and \quad N=\frac{1}{xy^2}-\frac{1}{y}$

$\therefore$ solution is $\int_{y\ constant} Mdx+\int N(term\ free\ from\ x)dy=c$

$$or,\ \int_{y\ const}\left(\frac{1}{x^2y}+\frac{2}{x}\right)dx-\int\frac{1}{y}dy=c \qquad or,\ -\frac{1}{xy}+2\ln x-\ln y=\ln c \quad or,\ -\frac{1}{xy}+\ln\frac{x^2}{y}=\ln c$$

This is the required solution

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Ex: Solve $(x^4+y^4)dx - xy^3dy = 0$

Solu: Given equation is

$(x^4+y^4)dx - xy^3dy = 0$ .....(i)

Here

$M = x^4+y^4 \qquad \therefore \frac{\partial M}{\partial y} = 4y^3 \qquad N = xy^3 \qquad \therefore \frac{\partial N}{\partial x} = y^3$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

Now,

$$Mx + Ny = (x^4+y^4)x - (xy^3)y = x^5 + xy^4 - xy^4 = x^5$$

$$\therefore IF = \frac{1}{x^5}$$

$$(i)\times IF \Rightarrow \frac{x^4+y^4}{x^5}dx - \frac{xy^3}{x^5}dy = 0 \quad or, \quad \left(\frac{1}{x}+\frac{y^4}{x^5}\right)dx - \frac{y^3}{x^4}dy = 0$$

$\therefore$ solution is $\int_{y\ const} Mdx + \int N(term\ free\ from\ x)dy = c$

$$or, \int_{y\ const}\left(\frac{1}{x}+\frac{y^4}{x^5}\right)dx + \int 0dy = c \qquad or, \quad \ln x + y^4.\frac{x^{-4}}{-4} = c \qquad or, \quad \ln x - \frac{y^4}{4x^4} = c$$

This is the required solution

Ex: Solve $(x^2+y^2+1)dx - 2xydy = 0$

Solu: Given equation is

$(x^2+y^2+1)dx - 2xydy = 0$ .....(i)

Here

$M = x^2+y^2+1 \qquad \therefore \frac{\partial M}{\partial y} = 2y \qquad N = -2xy \qquad \therefore \frac{\partial N}{\partial x} = -2y$

Since $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{-2xy}(2y+2y) = \frac{4y}{-2xy} = -\frac{2}{x} = f(x)$$

$$\therefore IF = e^{\int \frac{-2}{x}dx} = e^{-2\ln x} = x^{-2}\ \frac{1}{x^2}$$

$$(i)\times IF \Rightarrow \frac{x^2+y^2+1}{x^2}dx - \frac{2xy}{x^2}dy = 0 \qquad or, \quad \left(1+\frac{y^2}{x^2}+\frac{1}{x^2}\right)dx - \frac{2y}{x}dy = 0$$

$\therefore$ solution is $\int_{y\ const} Mdx + \int N(term\ free\ from\ x)dy = c$

$$or, \int_{y\ const}\left(1+\frac{y^2}{x^2}+\frac{1}{x^2}\right)dx + \int 0dy = c \quad or, \quad x + y^2.\frac{x^{-1}}{-1} + \frac{x^{-1}}{-1} = c \quad or, \quad x - \frac{y^2}{x} - \frac{1}{x} = c$$

$$or, \quad x^2 - y^2 - 1 = cx$$

This is the required solution

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Ex: Solve $(x^3-2y^2)dx+2xydy=0$

Solu: Given equation is

$(x^3-2y^2)dx+2xydy=0$ .....(i)

Here

$M=x^3-2y^2 \qquad \therefore \frac{\partial M}{\partial y}=-4y \qquad N=2xy \qquad \therefore \frac{\partial N}{\partial x}=2y$

Since $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

Now,

$$\frac{1}{N}\left(\frac{\partial M}{\partial y}-\frac{\partial N}{\partial x}\right)=\frac{1}{2xy}(-4y-2y)=\frac{-6y}{2xy}=-\frac{3}{x}=f(x)$$

$$\therefore\ IF=e^{\int\frac{-3}{x}dx}=e^{\ln x^{-3}}=x^{-3}=\frac{1}{x^3}$$

$$(i)\times IF\Rightarrow \quad \frac{x^3-2y^2}{x^3}dx-\frac{2xy}{x^3}dy=0 \quad or,\ \left(1-\frac{2y^2}{x^3}\right)dx+\frac{2y}{x^2}dy=0$$

$\therefore$ solution is $\int_{y\ constant} Mdx+\int N(term\ free\ from\ x)dy=c$

$$or,\ \int_{y\ const}\left(1-\frac{2y^2}{x^3}\right)dx+\int 0dy=c \quad or,\ x-\frac{2y^2}{-2x^2}=c \quad or,\ x+\frac{y^2}{x^2}=c \quad or,\ x^3+y^2=cx^2$$

This is the required solution

Ex: Solve $(3x^2y^4+2xy)dx+(2x^3y^3-x^2)dy=0$

Solu: Given equation is

$(3x^2y^4+2xy)dx+(2x^3y^3-x^2)dy=0$ .....(i)

Here

$M=3x^2y^4+2xy \qquad \therefore \frac{\partial M}{\partial y}=12x^2y^3+2x \qquad N=2x^3y^3-x^2 \qquad \therefore \frac{\partial N}{\partial x}=6x^2y^3-2x$

Since $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

Now,

$$\frac{1}{M}\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)=\frac{1}{3x^2y^4+2xy}(-2x(3xy^3+2)=\frac{-2x(3xy^3+2}{xy(3xy^3+2)}=-\frac{2}{y}=f(y)$$

$$\therefore\ IF=e^{\int\frac{-2}{y}dy}=e^{-2\ln y}=e^{\ln y^{-2}}=y^{-2}=\frac{1}{y^2}$$

$$(i)\times IF\Rightarrow \quad \frac{3x^2y^4+2xy}{y^2}dx-\frac{2x^3y^3-x^2}{y^2}dy=0 \quad or,\ \left(3x^2y^2+\frac{2x}{y}\right)dx+\left(2x^3y-\frac{x^2}{y^2}\right)dy=0$$

$\therefore$ solution is $\int_{y\ constant} Mdx+\int N(term\ free\ from\ x)dy=c$

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$$or, \int_{y\ const}\left(3x^2y^2+\frac{2x}{y}\right)dx+\int 0dy=c \qquad or, \quad \frac{3x^3y^2}{3}+\frac{2x^2}{2y}=c \qquad or, \quad x^3y^2+\frac{x^2}{y}=c$$

$$or, \quad x^3y^3+x^2=cy$$

This is the required solution

Ex: Solve $(y^4+2y)dx+(xy^3+2y^4-4x)dy=0$

Solu: Given equation is

$(y^4+2y)dx+(xy^3+2y^4-4x)dy=0$ .....(i)

Here

$$M=y^4+2y \qquad \therefore \frac{\partial M}{\partial y}=4y^3+2 \qquad N=xy^3+2y^4-4x \qquad \therefore \frac{\partial N}{\partial x}=y^3-4$$

Since $\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$ so the differential equation is not exact. But M and N are homogeneous.

Now,

$$\frac{1}{M}\left(\frac{\partial N}{\partial x}-\frac{\partial M}{\partial y}\right)=\frac{1}{y^4+2y}(y^3-4-4y^3-2)=\frac{1}{y(y^3+2)}(-3y^3-6)=\frac{-3(y^3+2)}{y(y^3+2)}=\frac{-3}{y}=f(y)$$

$$\therefore \ IF=e^{\int\frac{-3}{y}dy}=e^{-3\ln y}=e^{\ln y^{-3}}=y^{-3}=\frac{1}{y^3}$$

$$(i)\times IF \Rightarrow \quad \frac{y^4+2y}{y^3}dx-\frac{xy^3+2y^4-4x}{y^3}dy=0 \qquad or, \quad \left(y+\frac{2}{y^2}\right)dx+\left(x+2y-\frac{4x}{y^3}\right)dy=0$$

$\therefore$ solution is $\int_{y\ constant} Mdx+\int N(term\ free\ from\ x)dy=c$

$$or, \int_{y\ const}\left(y+\frac{2}{y^2}\right)dx+\int 2ydy=c \qquad or, \quad xy+\frac{2x}{y^2}+y^2=c$$

This is the required solution

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